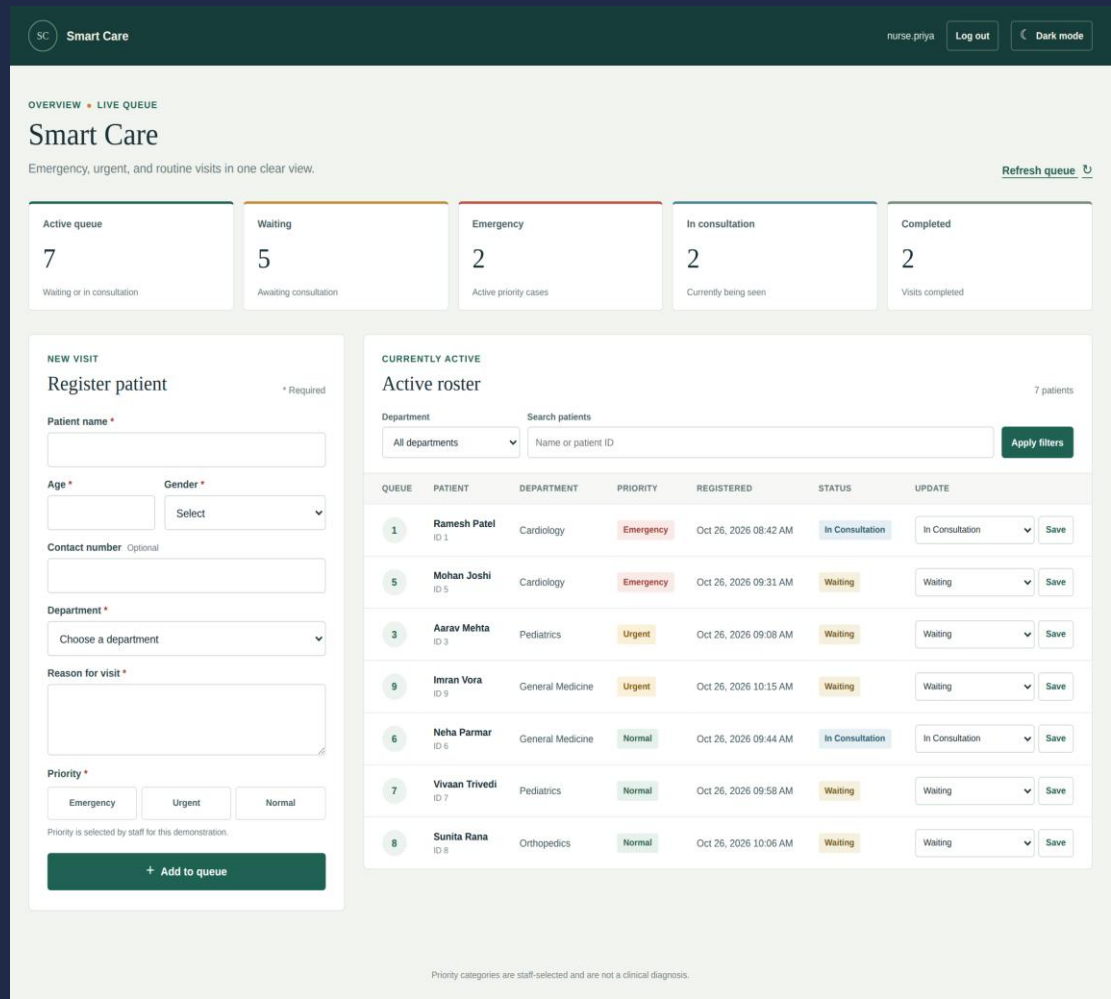
A. D. PATEL INSTITUTE OF TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY

Smart Patient Queue System

Emergency prioritization for hospital staff

Presented by: Savani Ayushi (12402080601014)
Branch : IT-A (Sem 5)



Manual queues slow hospitals down



Long, unorganized queues

Patients wait in the order they arrive, with no clear system.



Emergencies hard to spot

Critical cases are not identified quickly among the crowd.



Records kept by hand

Patient details on paper or memory get lost or mixed up.



Slow to find information

Searching for one patient's details takes too much time.

THE MAIN PROBLEM

No single place to register, prioritize and track patients.

No central dashboard for staff.

One staff dashboard for the whole queue



Objectives

- **Cut manual paperwork**
- **Serve emergencies first**
- **Give every patient a unique queue number**
- **Search by name or ID in seconds**
- **Track status until completed**

What makes it different



Secure staff login

Registration, login and protected pages



Patient registration

Name, age, gender, department, symptoms



3 priority levels

Emergency, Urgent, Normal with color badges



Auto queue number

Unique number, never repeated



Department filter

Show one department's queue



Smart search

Partial name or patient ID



Light and dark mode

Comfortable for any shift

How the priority queue works

ARRIVAL ORDER

Aarav Mehta 09:08 AM	Urgent
Mohan Joshi 09:31 AM	Emergency
Vivaan Trivedi 09:58 AM	Normal
Imran Vora 10:15 AM	Urgent

QUEUE ORDER

Mohan Joshi 09:31 AM	Emergency
Aarav Mehta 09:08 AM	Urgent
Imran Vora 10:15 AM	Urgent
Vivaan Trivedi 09:58 AM	Normal

SQL ORDER BY (simplified)

```
ORDER BY CASE priority_level
  WHEN 'Emergency' THEN 1
  WHEN 'Urgent' THEN 2
  ELSE 3
END,
registered_at ASC,
patient_id ASC;
```

Mohan arrived later than Aarav but is called first, because Emergency outranks Urgent. Equal priorities go in arrival order.

Priority is chosen by staff. The system organizes the queue; it does not diagnose patients.

System architecture: simple and secure

Hospital staff

Uses any browser

Web interface

HTML, CSS, JavaScript

Flask backend

Routes, sessions, validation

Jinja2 templates

Dynamic HTML pages

MySQL

patients, staff_accounts

Runs locally

Python + Flask + MySQL on a normal PC

Lightweight

Plain HTML, CSS and JS on the front end

Safe data access

Parameterized SQL stops SQL injection

Database design: two focused tables

patients

Column	Type	Key
patient_id	BIGINT, auto-increment	**PK**
patient_name	VARCHAR(100)	
age	TINYINT (max 120)	
gender / contact	VARCHAR(30)	
department	VARCHAR(100)	
symptoms	TEXT	
priority_level	ENUM: Emergency, Urgent, Normal	
queue_number	BIGINT	**UNIQUE**
status	ENUM: Waiting, In Consultation, Completed	
registered_at	TIMESTAMP, default now	

staff_accounts

Column	Key
staff_id	**PK**
username	**UNIQUE**
email	**UNIQUE**
password_hash	
created_at	

DESIGN NOTES

- ENUM allows only valid priority and status
- queue_number is UNIQUE, so it never repeats
- Index on status, priority and time speeds up the queue query

Technology stack and security

Layer	Technology
Frontend	HTML5, CSS3, JavaScript
Backend	Python, Flask
Templates	Jinja2
Database	MySQL
DB access	MySQL Connector (parameterized)
Authentication	Sessions + password hashing
Configuration	python-dotenv (.env file)

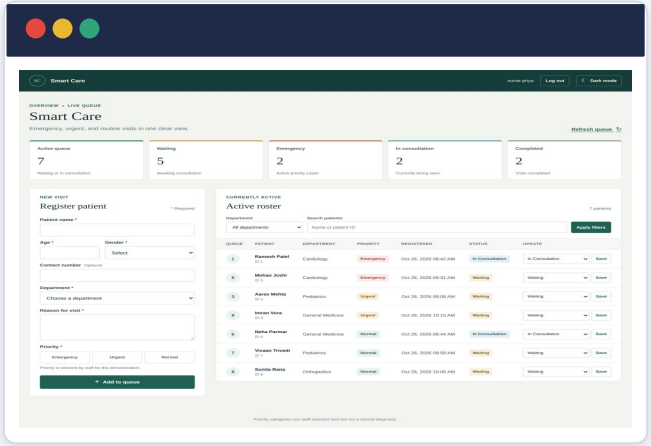
Security built in

- **Password hashing**
- **Parameterized SQL**
- **CSRF tokens**
- **Browser + server validation**
- **Login required for pages**
- **HttpOnly, SameSite cookies**

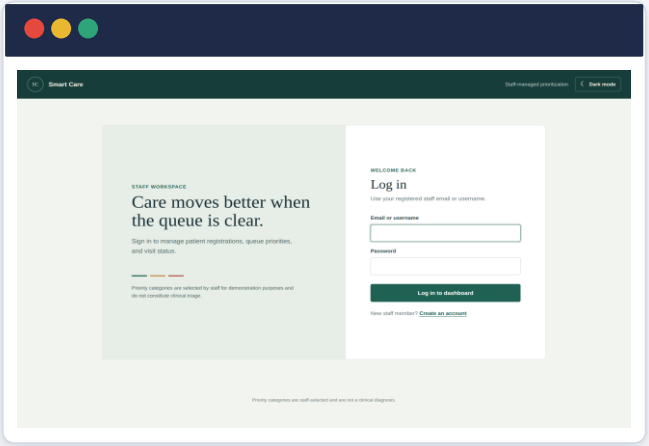
Demonstration project: do not enter real patient data.

How it works: from login to completion

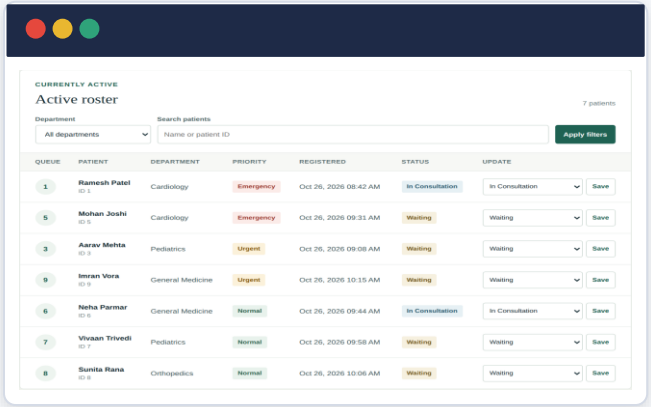
1. Staff log in with a secure session
2. Register patient with details and priority
3. Patient gets an automatic queue number
4. Dashboard orders queue: Emergency first
5. Search or filter by department, open details
6. Update status until Completed



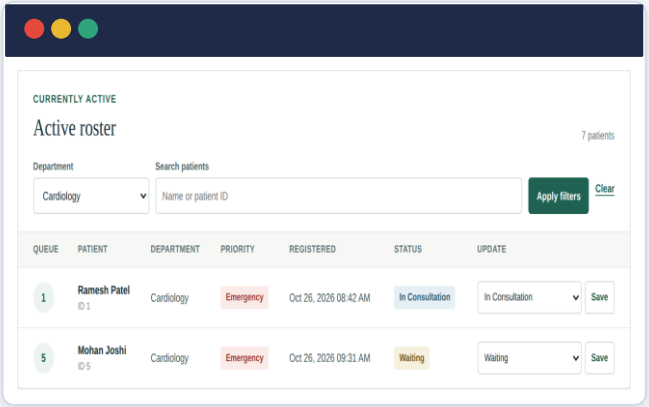
Dashboard



Secure login



Live queue



Department filter

Project Evaluation & Future Prospects

Advantages

- Faster patient registration
- Emergency cases stand out
- Easy name and ID search
- Simple status tracking
- Less manual work

Limitations

- Single hospital, demo use
- No account approval
- No rate limiting or audit log
- No notifications yet

Future scope

- Doctor login and roles
- SMS / WhatsApp alerts
- Waiting-time estimate
- Appointments and analytics
- Cloud deployment

Conclusion: The Smart Patient Queue System makes hospital queues faster, clearer and better organized. Thank you!